

Class 10 • Full Practice Test (Mathematics and Science)

Time: 2½ hours • Maximum marks: 69 • Answer all questions. Show your working.

Section A • Mathematics

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| Q1. | The LCM of 12 and 18 is:
(a) 36 (b) 72 (c) 6 (d) 216 | [1] |
| Q2. | Find the zeroes of the polynomial $x^2 - 5x + 6$ and verify the relationship between the zeroes and the coefficients (sum of zeroes). | [2] |
| Q3. | Solve by the elimination method: $x + y = 7$ and $x - y = 1$. | [3] |
| Q4. | Assertion (A): The equation $x^2 + 4 = 0$ has no real roots.
Reason (R): If the discriminant of a quadratic equation is negative, it has no real roots.
(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. | [1] |
| Q5. | Find the 10th term of the AP: 2, 7, 12, ... | [2] |
| Q6. | In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm, find EC. | [3] |
| Q7. | Find the distance between the points (3, 4) and (-1, 1). | [2] |
| Q8. | Prove that $(1 - \cos^2 A) \cdot \operatorname{cosec}^2 A = 1$. | [3] |
| Q9. | A tower casts a shadow 20 m long on level ground when the Sun's angle of elevation is 60° . Find the height of the tower. | [3] |
| Q10. | A point A is 5 cm from the centre O of a circle of radius 3 cm. Find the length of the tangent AP drawn from A to the circle. | [2] |
| Q11. | Find the area of a sector of a circle of radius 7 cm with central angle 60° . (Use $\pi = 22/7$.) | [3] |
| Q12. | A solid toy is in the form of a hemisphere of radius 3.5 cm surmounted by a right circular cone of the same radius and height 12 cm. Find the volume of the toy. (Use $\pi = 22/7$.) | [5] |
| Q13. | Find the mean of the following data:
x: 2, 4, 6, 8 f: 3, 5, 4, 2 | [3] |
| Q14. | A die is thrown once. Find the probability of getting (i) a prime number [2]; (ii) a number greater than 6 [1]. | [3] |

Section B • Science

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|------|---|-----|
| Q15. | Write the balanced chemical equation for the reaction of sodium with water and name the type of reaction. | [2] |
| Q16. | The pH of a neutral solution at room temperature is:
(a) 0 (b) 7 (c) 14 (d) 1 | [1] |

Q17.	Write the balanced chemical equation for the reaction of zinc with dilute hydrochloric acid [2]. How will you test the gas evolved? [1]	[3]
Q18.	Why does carbon form compounds mainly by covalent bonding?	[2]
Q19.	Name the tube that carries food from the mouth to the stomach, and the process by which food moves along it.	[2]
Q20.	What is a reflex action? Trace the path of a reflex arc.	[3]
Q21.	In yeast, asexual reproduction takes place mainly by: (a) budding (b) binary fission (c) fragmentation (d) spore formation	[1]
Q22.	A pure tall pea plant (TT) is crossed with a pure dwarf pea plant (tt). What will be the phenotypic ratio [1½] and the genotypic ratio [1½] in the F ₂ generation?	[3]
Q23.	(a) An object is placed 10 cm in front of a convex lens of focal length 15 cm. Find the position and nature of the image. [3] (b) Draw the ray diagram for this case. [2]	[5]
Q24.	Why does the clear sky appear blue?	[2]
Q25.	A copper wire is 2 m long and has an area of cross-section $1 \times 10^{-6} \text{ m}^2$. The resistivity of copper is $1.6 \times 10^{-8} \Omega \text{ m}$. Find its resistance.	[3]
Q26.	State Fleming's left-hand rule.	[2]
Q27.	Case-based: In a grassland, the food chain is: grass → grasshopper → frog → snake. (i) At which trophic level is the frog? [1] (ii) If the grass traps 10,000 J of energy, how much energy reaches the snake (10% law)? [1] (iii) What will happen to the grasshopper population if all the frogs are removed? Give a reason. [2]	[4]

Q28.

[1 marks]

The middle term of the A.P.: 10, 7, 4, ..., -62 is:

- (A) -26
- (B) -29
- (C) -32
- (D) -35

Q29.

[1 marks]

Assertion (A): In an ecosystem, 1% of energy is available for transfer from one trophic level to the next.

Reason (R): An average of 10% of the food eaten by organisms is available for flow in the form of chemical energy.

- (A) Both A and R are true, and R is the correct explanation of A.
- (B) Both A and R are true, and R is not the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Q30.

[2 marks]

Find the area of a sector of a circle with diameter 28 cm, if the length of the corresponding arc is 22 cm.

Q31.

[2 marks]

Heat produced per second due to a current in a resistor of $4\ \Omega$ is 400 joules. Calculate the potential difference across the resistor.

Q32.

[3 marks]

In what ratio does the x-axis divide the line segment joining the points $(-4, -6)$ and $(-1, 7)$? Also, find the coordinates of the point of division.

Q33.

[3 marks]

The lowest part of the ear, called earlobe, is closely attached to the side of the head in some of us (attached earlobe), and not in others (free earlobe). Attached and free earlobes are two variants found in human populations. The gene for free earlobe is dominant over attached earlobes.

- (a) A man with attached earlobes marries a woman having free earlobes. 50% of their children have free earlobes and 50% have attached earlobes. Explain the inheritance of this trait and write the trait combinations of the progeny. [2]
- (b) Write the gene combinations of the father and the mother in the above case. [1]

Q34.

[3 marks]

A hormone 'X' is secreted in blood when a person is under scary situation.

- (a) Identify the hormone 'X' and the gland that secretes it. [1]
- (b) Explain its role in dealing with scary or emergency situations. [2]

Q35.

[5 marks]

Tara scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each wrong answer, then Tara would have scored 50 marks. Assuming that Tara attempted all questions, find the total number of questions in the test.

Q36.

[4 marks]

Case study: A school conducted a weekly test for Class X students, before the commencement of the pre-board examination and recorded their scores (out of 50). To analyse performance patterns, the academic coordinator grouped the marks into intervals. The grouped frequency distribution is as below:

Marks obtained | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50

Number of students | 3 | 6 | 12 | 15 | 14

Based on the above information, answer the following questions:

- (i) Identify the class with the most common performance range. [1]
- (ii) Find the class interval containing the median. [1]
- (iii) What is the average performance of the students? [2]

Q37.

[2 marks]

A. Due to narrowing of the bile duct, the bile secretion from the liver to the small intestine was obstructed. How will the process of digestion be affected? [1]

B. How does the length of the intestine depend on the nature of food habit in different animals? [1]

Q38.

[2 marks]

A hall has a length of 9.75 m, breadth of 6.75 m and height of 5.25 m. What is the length of the longest unmarked ruler that can exactly measure the dimensions of the hall?